

Eye Care: Mobile Based Solution For Eye Illness Detection

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Abstract - Eye Care is a mobile based application developed as a solution for problems associated with eye illnesses. People are running out of their time and they do not check their health status daily. As a result of this situation people are suffering from diseases which they even do not know. There is no restriction of using this application by anyone for identifying eye illnesses. Furthermore any person can use this application without relying on a third party. "Eye Care" is the use of Information and Communication Technologies (ICT) for health sector specially to identify the preliminary levels of eye illnesses namely Cataract, Blepharitis, Chalazion and Conjunctivitis. The uniqueness of this research is, it identifies four main eye illnesses which are prominent at present with the use of image processing and semantic web techniques. "Eye Care" identifies above illnesses accurately and efficiently while providing extra knowledge about the illness to the user. Anyone can easily use this application without any challenge as well as a user can operate this android application easily without a single help from a third party.

Key words – Cataract, Conjunctivitis, Blepharitis, Chalazion, Image processing, Semantic Web

I. INTRODUCTION

At present eye illnesses have become prominent due to the negligence of people. People are hesitant to wait in the queues of hospitals since they are busy with their daily schedule. The commonly and largely used solution in this regard is home remedies. Still some home remedies are associated with many disadvantages. When a person takes home remedies for eye illnesses he/she should wait for a longer period of time to cure and since the person does not know the correct dosage can lead to an excessive ingestion. Previous researches in this

field have led to the invention of several mobile based applications. However none of them were provided a complete solution for the eye illnesses. Eye Test app [1], Eye Care Plus app [2], Vision Test 2.0 app [3], EyeXam app [4], Visual acuity Test app [5], White eye detector app [6] and Pink eye [7] app are some of the eye related applications where it is mainly focusing on eye exercise. Pink eye app [7] gives symptoms of Conjunctivitis but not going to detect illnesses.

Eye Xam [4] application provides eye exercises but not detecting any eye illness.

In this research paper we present the “Eye Care” mobile based application which detects main four common eye illnesses called Blepharitis [8], Conjunctivitis [9], Chalazion [10] and Cataract [11]. Yet it will overcome the problems of existing mobile based applications and enable the user to use the application by him. The user will no longer have to wait in the queues in the hospitals or try home remedies since Eye Care application facilitates user to check eye illnesses by his/her own.

User can operate the application without getting support from a third party. Eye Care application provides user to find eye consultants and provides immediate home precautions. Since the application does not take any private information of the user, he/she can use this application without any fear. This application enables the user to capture images as well as to choose images from gallery.

To detect eye illnesses the application must identify eyes of the user. Every face has distinguishable landmarks, different peaks and valleys that make up facial features. Human face approximately consisting of 80 nodal points such as width of the nose, depth of eye sockets, shape of cheek bones and the distance between two eyes.

Eye Care mobile based application figures out user’s face, left eye and right eye. In order to detect eye illnesses Eye Care application uses semantic web and image processing techniques to get accurate results. This system is designed not only for eye patients. It can use by any user who wish to check eyes.

Illnesses which are detected by eye applications are Conjunctivitis, Cataract, Chalazion and Blepharitis. Conjunctivitis which is also known as pink eye is an inflammation of the conjunctiva. Blepharitis is a common eye condition that causes inflammation of the eye lid. Chalazion is a small bump that appears on the eye lid because of a blocked oil gland. Cataract is a cloud of the eye lens that affects the vision.

Summary of the survey

As in the survey all android based applications address only eye exercises [1]. There couldn’t find any application which detects eye illnesses. To overcome those problems Eye Care eye illnesses detecting application is introduced which is designed to identify most prominent four eye illnesses. This application is designed in an attractive way so that patients can interact with the application very easily.

II. METHODOLOGY

The intention of this research project is to function as an eye detecting application. In order to detect eye illnesses image processing and semantic web techniques have used. The entire procedure will be sub divided into four modules as Image preprocessing and segmentation, data extraction and segmentation, Creating rules by knowledge base and Finalizing output by the results gained by data extraction and knowledge base.

- Image preprocessing and segmentation

First Eye Care android application facilitates user to enter an image with the help of camera or else user can choose an image from the gallery which is in the android phone. Eye Care application supports user by displaying the distance in the phone.

Energy of light around the user is measured by this application by using the light sensor available in the phone which avoids user to capture images in darker places. In order to get a high quality image user must need to maintain the distance between user and the android phone. Then image is uploading to the server to detect eye illnesses. Apache tomcat V8.0 is used as a server.

First task of the server application is to detect and crop eyes and irises from the image received by android application. In order to detect and crop eyes haarcascade files were used. There are no haarcascade files available to detect and crop irises. By using Haar training stuff tool, Eye Care has trained a haarcascade file to detect and crop irises. Cropped irises and eyes are then sending to extract relevant data to identify eye illnesses separately. Figure 1 show how it detected and cropped eyes by suing haarcascade files.



Figure 1: Eye Detection and cropping

- Data Extraction and Classification

Server has chosen as a java dynamic web project. Image which is captured by a user contains several noises. One of them is camera flash effect. Below mentioned figure shows a camera flash effect image.



Figure 2: Camera Flash Effect

In order to remove this camera flash effect noise, each and every pixel information was note down by using a pixel finder tool. Maximum pixel values of the camera flash effect are mentioned below with the aid of a table.

Table 1: Pixel Values

Pixel	Value
Red	230
Green	223
Blue	216

If the pixels are more than the above range it is considered as a camera flash effect pixel. These values are found by using RGB color model.

Conjunctivitis which is the redness of the eye is analyzed by using HSB color model. An equation must be used in order to detect Conjunctivitis.

$$P=H*360 \quad (1)$$

H is the extracted hue pixel and P is the new pixel. If P is between the range of 330 and 360 it can conclude that the pixel is a Conjunctivitis affected pixel.

Cataract which means the white cloud of the iris was identified by suing RGB color model. Every single pixel was note down by using pixel founder tool. Below mentioned are the values gained by the pixel founder tool related to Cataract illness.

Table 2: Pixel values of Cataract

Pixel	Value	
	Min	Max
Red	100	170
Green	100	175
Blue	100	195

If the pixel value is between the above min and max range it can conclude that the pixel is a cataract affected pixel. After detecting the illness then the results are sending to knowledge base in order to detect the illness further.

- Creating knowledge Base

The implementation of ontology was made relying on the framework based on knowledge and ontology editor called Protégé. The ontology was encoded using the Web Ontology Language (OWL), which provides classes, properties, individuals and data type values. When creating the knowledge base it uses the data and get from the data extraction process. Here it is training the system dynamically to identify the illnesses. Many applications require the ability to work with well-defined portions of ontology, which are usually

called modules. The OWL API contains basic functionality for extracting syntactic-locality based modules [12] from ontologies. These kinds of modules are based on the notion of conservative-extensions [13], and guarantee to preserve entailments from ontology over a given signature. As an initial step first creates the ontology and then updates it dynamically.

To build the ontology, we gathered data that extracted from image processing. Analyzed hundreds of images to come up with a conclusion that in which range is belong to inflected eye or not. After then we came up with the final ranges and then added those values into ontology. After built the ontology statically then have to update it in a dynamic way. To do it, it must load the existing ontology into our dynamic web project and implemented the code.

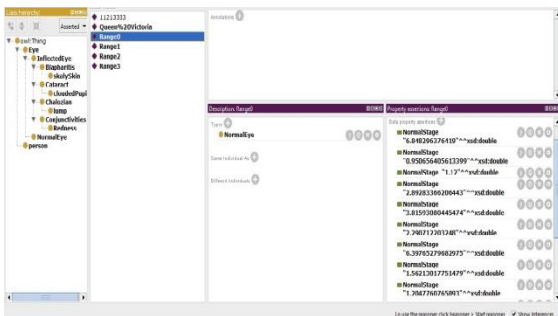


Figure 3: Ontology class structure

Table 3: Ranges of conjunctivitis

Stage of the conjunctivitis	Range (%)
Normal stage	0-7
Initial stage	7.1-10.00
Average stage	10.1-17.5
Critical/Severe stage	Above 17.6

- Finalizing output

In component Data analyzing and finalize output, it matches the input data with the knowledgebase rules and then writes SPARQL select queries to retrieve data. According to the input value which gained by data extraction, here it creates the select queries for the knowledge base to retrieve class name for a given individual value. While this process is handling in the server, android application prompts questions to the user to answer them thoroughly.

Those questionnaires belong to four illness categories name as Conjunctivitis, Chalazion, Blepharitis and Cataract. These questions mainly based on some of the major symptoms of above illnesses. By analyzing answers provided by the user then decides the eye condition of the user.

According to the answers it calculates a percentage value of having the illness. These percentage values categorize in to four stages as Normal stage, Average stage, Critical stage and severe stage.

In server it retrieves the illness name from the knowledge base and by using a servlet it sends the result as a signal to the android application. Finally it gives the eye condition by analyzing answers of the questionnaires and output data retrieve from the knowledge base. And then displays the output to the user. It gives a brief description about the illness and the details about the consultants and hospitals near the user.

III. RESEARCH FINDINGS

There are numerous Haar cascade files to detect eyes, faces and various objects. But still researchers have failed to develop a Haar cascade file to detect iris in java language. In “Eye Care” application it has developed a Haar cascade file for iris detection.

Measuring the light source option helps the client to get a quality image with recommended light energy which uses smart phone light sensor to measure the light energy. If the light energy is lower than the recommended range (1000lu) android phone is not allowing capturing the image.

In distance calibration the distance between user and the android application must be calculated in order to maintain the quality of the image. Here the android phone will guide the user to set the distance correctly. When user zoom the camera, it changes the distance between two eyes. When the distance is satisfied user can capture the image.

IV. DISCUSSION

Eye Care application detects four main eye illnesses called Conjunctivitis, Blepharitis, Cataract and Chalazion by using image processing and semantic web techniques.

First the input image is segmented for irises and eyes and then it extracts relevant data. Extracted data are sending to the knowledge base for further processing and then it finalize the output with the aid of results gained by knowledge base, extracted data and the answers provided by the user.

Following is the accuracy rate of the Eye Care application which was gained by testing 150 images. This is considered as system testing.

No of input images: 150

No of correct results: 125

Accuracy = $(125/150) \times 100\%$

= 83.33%

According to the above answer it can be concluded that the accuracy of the Eye Care system is about 83.33%.

V. CONCLUSION

In this paper we present the development of the eye detecting application, which we believe is the ideal solution for eye illnesses patients. We proposed an android based mobile application where it identifies

most prominent eye illnesses with the use of image processing and semantic web techniques. Eye Care application provides the users with various benefits over existing applications which are available in the market. Those who use this application they can identify the eye illnesses within a short period of time with the aid of android phone and generate results including home precautions that he/she can take to reduce the illness up to some extent and finally they can consult a doctor to cure the illness completely.

There are considerable amount of eye related mobile applications are developed in the field of information technology. Our approach is completely automatic and optimized solution comparing with all others existing. Eye Care application allows user to operate it by his own, without relying on others support. The only limitation of this application is that the Eye Care application detects only four illnesses.

VI. FUTURE WORK

With the use of image processing techniques it detects and crop irises of the provided image. In iris detection and iris cropping it detects and crop irises accurately with some of the garbage values.

Here the system cannot identify the iris directly. When user checks the image folder he/she can view the irises photos clearly. So this is one of the limitations of this application. In order to overcome

the above mentioned problems developer can train a Haar cascade file where it needs about 60 hours to create the Haar cascade file. User must enter about 2000 images of positive images and 5000 images of negative images in order to create a better Haar cascade file.

Here quality of the picture is an issue. If user captures an image from a poor quality camera it is hard to identify iris and eyes in the server application. Therefore in order to minimize the above problem developer must build new Haar cascade files or algorithms which supports for the low quality cameras.

If the image contains a group photo or else other images server is falling down since the application is detecting only two eyes in a human image. If someone wishes to improve Eye Care application he/she can find a way to detect eyes in a group photo. This application is only focusing on four eye illnesses. Therefore a person can improve the system which can detect several illnesses. Eye Care application support only for android phones. Therefore a developer can build Eye Care application which will support for several platforms.

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